Temporal Kolmogorov–Arnold Networks for High-Frequency Order Flow Imbalance Classification: A Cautionary Benchmark

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1 Introduction

The microstructure of modern electronic markets is governed by the Limit Order Book (LOB), in which the continuous arrival, modification, and cancellation of orders drives price formation. Among the many features extracted from LOB data, Order Flow Imbalance (OFI) — the net signed pressure of bid versus ask updates at the top of the book — has emerged as particularly informative. Cont et al. [1] showed that OFI bears a strong, *near-linear* relationship to contemporaneous mid-price changes, and subsequent work has confirmed its predictive value for short-horizon price movements [5]. Forecasting the next-step direction of OFI is therefore directly relevant to algorithmic execution and short-term alpha generation.

The task is statistically hostile: LOB data is non-stationary, heavy-tailed, and dominated by microstructure noise. Deep sequence models such as Long Short-Term Memory networks [4] and DeepLOB [8] have become standard tools for LOB modelling, but both rely on fixed activation functions and linear weights, which can require large parameter counts to fit highly non-linear surfaces and offer limited interpretability. Liu et al. [6] recently proposed Kolmogorov–Arnold Networks (KAN), which place *learnable* univariate B-splines on the edges of the network and simple sums at the nodes. This design is grounded in the Kolmogorov–Arnold representation theorem and has been shown to match or exceed MLPs at lower parameter counts while permitting direct visualisation of the learned functions. Genet and Inzirillo [2] extended this idea to sequential data with the Temporal KAN (TKAN), embedding KAN layers inside an LSTM-style gated cell with four KAN gates and a cell state.

This report benchmarks a minimal ungated recurrent KAN cell — differing from [2] in that it omits LSTM-style gating to isolate the spline contribution — against an LSTM and a multinomial logistic regression null baseline. The inclusion of a linear null is motivated by methodological critiques showing that omitted linear benchmarks inflate the apparent contribution of deep architectures in applied financial ML [3, 7].

2 Methodology

2.1 Data and feature engineering

We use Level-2 LOB data for AAPL (NASDAQ) over 21 consecutive trading days from 22 September to 20 October 2025, sourced from LOBSTER message and order-book files, containing approximately 19.2 million tick-level events. Top-2 bid and ask price and size levels are aggregated from raw events to a 10-second resolution by summing OFI events and averaging book-state quantities, yielding 47,403 samples after feature construction. The 10-second resolution was a compromise between

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signal preservation and computational tractability given the walk-forward training budget; we discuss the implications of this choice in Section 4.

Following Cont et al. [1], for each bid and ask event n in the time bin ending at t we define the signed contribution:

$$e_n^b = \begin{cases} V_t^b & \text{if } P_t^b > P_{t-1}^b \\ V_t^b - V_{t-1}^b & \text{if } P_t^b = P_{t-1}^b \\ -V_{t-1}^b & \text{if } P_t^b < P_{t-1}^b \end{cases}, \qquad e_n^a = \begin{cases} V_t^a & \text{if } P_t^a < P_{t-1}^a \\ V_t^a - V_{t-1}^a & \text{if } P_t^a = P_{t-1}^a \\ -V_{t-1}^a & \text{if } P_t^a > P_{t-1}^a \end{cases} \tag{1}$$

$$\text{OFI}_t \;=\; \sum_{n \in \text{bin}_t} \left(e_n^b - e_n^a\right) \tag{2}$$

To enforce stationarity, raw absolute prices are removed and replaced with relative features: top-1 and top-2 distances from the mid-price, top-1 and top-2 volume imbalances $(V^b - V^a)/(V^b + V^a)$, the micro-price pressure $P^{\text{micro}} - P^{\text{mid}}$, and rolling 60-step OFI momentum and mid-price-return volatility. The final feature vector has dimension 15. The target is the directional regime of OFI_{t+1}, encoded as Strong Sell (bottom 20%), Neutral (middle 60%), or Strong Buy (top 20%), with quantile thresholds recomputed from training data only at every fold.

2.2 Ungated recurrent KAN cell

A KAN layer transforms an input vector $\mathbf{x} \in \mathbb{R}^d$ through learnable univariate functions on each edge [6]:

$$y_o \;=\; \sum_{i=1}^{d} \phi_{o,i}(x_i), \qquad \phi(x) \;=\; w_b\, \text{SiLU}(x) \;+\; w_s \sum_{m=1}^{G+k} c_m\, B_m(x) \tag{3}$$

where B_m are B-spline basis functions of degree k defined recursively over a grid of $G+1$ knots via the Cox–de Boor recursion, and w_b, w_s, c_m are learnable parameters. The residual SiLU path stabilises training; the spline path provides the expressive non-linearity. Inputs are clamped via $\tanh$ to remain within the fixed spline grid $[-1, 1]$. We embed this in a minimal ungated recurrent cell:

$$\mathbf{h}_t \;=\; \tanh(\text{KAN}([\mathbf{x}_t, \mathbf{h}_{t-1}])) \tag{4}$$

The final hidden state is layer-normalised and passed to a linear classifier head producing three logits. Figure 1 contrasts this design with the standard LSTM cell and the original gated TKAN of Genet and Inzirillo [2].

2.3 Baselines

Logistic Regression (null). Multinomial logistic regression with `class_weight='balanced'` serves as the null baseline required by the coursework rubric. It receives the same 15 features at the current timestep and has no temporal context.

LSTM (peer). A single-layer LSTM with the same input dimension serves as a deep-learning peer comparison; the final hidden state is passed through a linear classifier.

2.4 Training and validation

All deep models are trained with class-weighted cross-entropy ($w = [3, 1, 3]$, label smoothing 0.1) and AdamW (learning rate 5×10^{-4}, weight decay 5×10^{-4}), with gradient clipping at norm 1.0 and early stopping (patience 2) on validation loss. Hyperparameters are selected by Optuna (TPE sampler, 20 trials, objective: maximise macro-F1 on a 3-day held-out window). The chosen TKAN configuration is hidden size 6, sequence length 9, grid size 3, spline order 3, two layers.

We use a **day-forward expanding-window** validation: at fold i, models train on days $1 \dots i$ and test on day $i+1$, yielding 15 folds. Quantile thresholds and `StandardScaler` parameters are refit on training data only at each fold. Sequence windows are strictly bounded by daily market closures to prevent overnight leakage. We report mean $\pm$ standard deviation across folds for accuracy and macro-F1, and use the paired Wilcoxon signed-rank test on per-fold macro-F1 to assess the significance of pairwise differences between models.

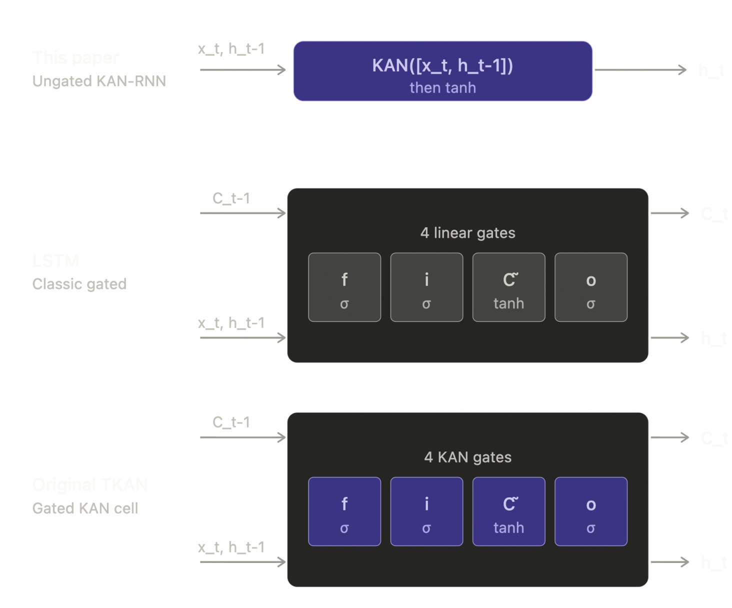


Figure 1: Recurrent cell architectures compared. Purple denotes KAN-based transformations (learnable B-spline edges); gray denotes standard linear weights. Our model uses a single ungated KAN transformation per step without a cell state; LSTM uses four linear gates with a cell state stream; the original TKAN of Genet and Inzirillo (2024) combines four KAN gates with LSTM-style gating.

3 Results

Table 1: Day-forward validation results, mean $\pm$ std over 15 folds. Best per column in bold.

Model	**Macro-F1**	**Accuracy**
Logistic Regression (null)	$\mathbf{0.419 \pm 0.017}$	$\mathbf{0.590 \pm 0.080}$
LSTM (peer)	0.411 ± 0.025	0.554 ± 0.084
TKAN (ungated)	0.402 ± 0.037	0.521 ± 0.092

Table 1 reports the final results across 15 walk-forward folds. The ranking is the opposite of our initial hypothesis: the logistic regression null baseline achieves the highest macro-F1, followed by LSTM, with TKAN last. Logistic regression also exhibits the lowest fold-to-fold variance (0.017), roughly half of TKAN's (0.037), indicating that its performance is not only higher on average but also more consistent across heterogeneous trading days.

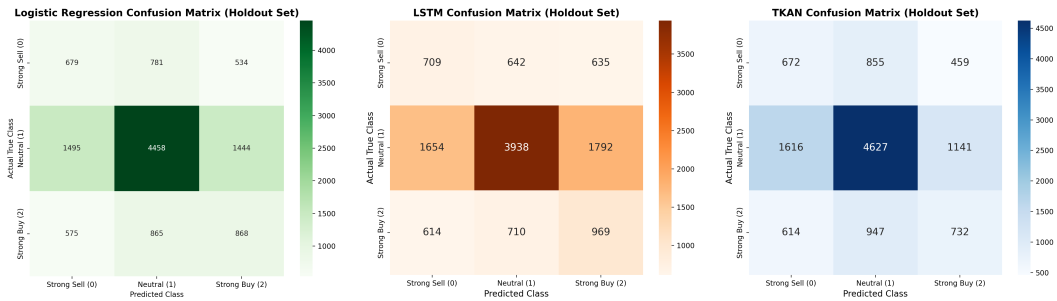
Table 2: Paired Wilcoxon signed-rank tests on per-fold macro-F1 ($n = 15$ folds).

Comparison	**Mean diff.**	**W statistic**	**p-value**
TKAN vs Logistic Regression	-0.017	25.0	0.048*
LSTM vs Logistic Regression	-0.008	23.0	0.035*
TKAN vs LSTM	-0.009	38.0	0.229

* significant at $\alpha = 0.05$.

Table 2 reports paired Wilcoxon tests on the 15 per-fold macro-F1 scores. Both deep models are significantly beaten by logistic regression. However, the difference between TKAN and LSTM is *not* statistically significant ($p = 0.23$): the two deep architectures perform equivalently on this task, and both lose to the linear baseline. The additional representational capacity of the deep models does not translate into predictive gains here.

Figure 2 shows confusion matrices on the final holdout period. All three models distribute predictions across the three classes rather than collapsing to the Neutral majority — a consequence of the class-weighted losses. Logistic regression, however, exhibits the cleanest diagonal structure on the two extreme classes, while the deep models exhibit greater confusion between Neutral and the extreme classes, suggesting they overfit noisy intra-day fluctuations.



(a) Logistic regression. (b) LSTM. (c) TKAN.

Figure 2: Confusion matrices on the holdout test period. All three models avoid collapsing to Neutral, but the deep models exhibit greater confusion between Neutral and the extreme classes.

A final and more positive observation concerns interpretability. Figure 3 plots the twelve most non-linear learned univariate functions $\phi_{o,i}$ inside the TKAN recurrent cell, selected by residual standard deviation after linear detrending. Several functions exhibit clear monotone, sigmoidal, or peaked shapes, demonstrating that the spline parametrisation did learn smooth, interpretable mappings from specific LOB features (e.g. `ofi`, `ask_dist_1`, `vol_imb_1`) into specific recurrent state dimensions. This form of direct transparency is not available from the LSTM baseline, and is preserved even when predictive performance does not surpass the linear null.

4 Discussion

The empirical result — a linear null baseline significantly beating both deep architectures, which are themselves statistically tied — initially appears disappointing, but we argue it is both interpretable and consistent with the broader empirical asset-pricing literature. We identify five plausible and non-mutually-exclusive factors.

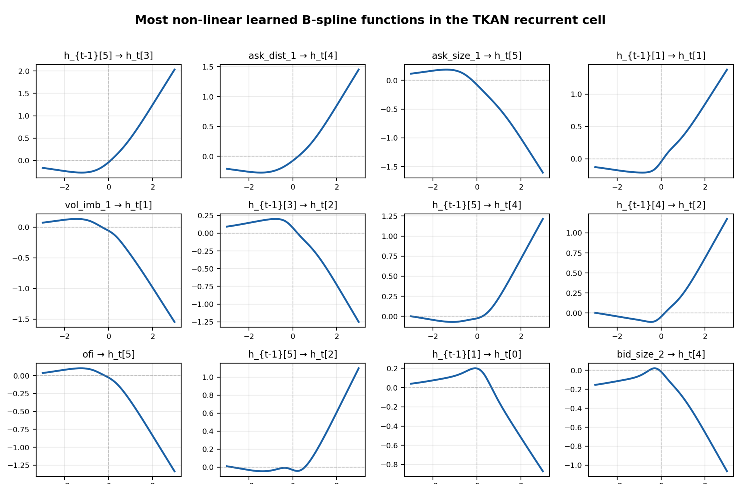


Figure 3: The twelve most non-linear learned B-spline functions in the TKAN recurrent cell, selected from the full 6×21 grid by residual standard deviation after linear detrending. Titles indicate the input coordinate (feature name or previous hidden dimension) and the output hidden dimension.

Near-linearity of the OFI–price relationship. Cont et al. [1] showed that OFI has an approximately linear relationship with contemporaneous mid-price changes. If the underlying structure being learned is near-linear, a well-specified linear model will be hard to beat: the residual non-linear signal available to a deep model is small, and any additional representational capacity primarily fits noise. Our task — predicting the directional regime of next-step OFI — inherits this near-linear character.

Temporal aggregation to 10 seconds discards sub-bin dynamics. The empirical OFI literature consistently finds that the predictive content of OFI is concentrated at horizons of seconds to a few tens of seconds, decaying rapidly beyond that [1, 5]. By aggregating tick data into 10-second bins — a choice forced on us by computational constraints in our walk-forward validation and Optuna search — we discard the sub-second dynamics that deep sequence models would be best positioned to exploit. At 10-second resolution, the remaining signal is dominated by coarse-grained microstructure state that is well summarised by the 15 static features we engineered, leaving little residual temporal structure for the LSTM or TKAN recurrences to model beyond what logistic regression already captures. We regard this as the single most likely explanation for the deep models' under-performance.

Absence of a gated cell and a persistent cell state. As illustrated in Figure 1, our TKAN is a minimal Elman-style recurrent cell with a single ungated `tanh` recurrence, in contrast to the four-gate LSTM with its separate cell state. The gated structure provides an explicit mechanism for preserving information across time and mitigating vanishing gradients, which our cell lacks. The gap between TKAN and LSTM (although not significant) and the larger gap to logistic regression may therefore reflect the absence of gating rather than a deficiency of the B-spline representation per se. A natural and important follow-up is to implement the full gated TKAN of Genet and Inzirillo [2], which combines KAN layers with LSTM-style gating, to disentangle these two effects. Our current design confounds "spline versus linear weights" with "ungated versus gated recurrence", so we cannot attribute the performance gap to either factor alone.Despite this predictive underperformance, the KAN function visualisation in Figure 3 demonstrates that interpretability is preserved: the spline parametrisation successfully captured smooth, varied, and directly visualisable univariate mappings, a degree of model transparency unavailable from either the LSTM or the logistic regression baseline.

Absence of grid adaptation. Our `KANLinear` layer uses a *static* knot grid fixed at $[-1, 1]$, with inputs clamped via $\tanh$ before spline evaluation. This design has two coupled weaknesses. First, $\tanh$ saturates for $|x| > 2$: any feature value in the tails of the standardised distribution is squashed into a narrow band near ± 1, and its gradient contribution to the spline coefficients is compressed by $\tanh'$, contributing to a mild vanishing-gradient effect on the tail samples where most of the informative OFI extremes live. Second, because the knots are fixed and uniformly spaced, knot intervals that receive very little data after the $\tanh$ squash (for most standardised features, the extreme ends of $[-1, 1]$) cannot develop meaningful spline coefficients — the fit is concentrated in whichever few intervals carry most of the mass. Liu et al. [6] explicitly address this in the original KAN paper through a *grid adaptation* procedure in which the knot positions are periodically re-calculated during training so that each knot interval contains roughly equal data mass, concentrating expressive capacity where the activations actually live. We did not implement grid adaptation, and expect it to have a materially positive effect.

Future work. The most important follow-up is to implement the full gated TKAN with KAN grid adaptation enabled, and to re-run the experiment at 1-second or event-driven resolution. These three changes jointly address the three factors we believe most plausibly explain the current gap. A second direction is to switch the target from next-step OFI to short-horizon mid-price returns, which are more directly meaningful for trading applications. A third is to extract symbolic rules from the learned splines, as proposed by Liu et al. [6], which would turn interpretability into concrete economic hypotheses. All of these extensions should be repeated across multiple stocks and longer time windows so that results are not contingent on a single instrument or market regime.

References

[1] R. Cont, A. Kukanov, and S. Stoikov. The price impact of order book events. *Journal of Financial Econometrics*, 12(1):47–88, 2014.

[2] R. Genet and H. Inzirillo. TKAN: Temporal Kolmogorov–Arnold networks. *arXiv preprint arXiv:2405.07344*, 2024.

[3] S. Gu, B. Kelly, and D. Xiu. Empirical asset pricing via machine learning. *The Review of Financial Studies*, 33(5):2223–2273, 2020.

[4] S. Hochreiter and J. Schmidhuber. Long short-term memory. *Neural Computation*, 9(8):1735–1780, 1997.

[5] P. N. Kolm, J. Turiel, and N. Westray. Deep order flow imbalance: Extracting alpha at multiple horizons from the limit order book. *Mathematical Finance*, 33(4):1044–1081, 2023.

[6] Z. Liu, Y. Wang, S. Vaidya, F. Ruehle, J. Halverson, M. Soljačić, T. Y. Hou, and M. Tegmark. KAN: Kolmogorov–Arnold networks. *arXiv preprint arXiv:2404.19756*, 2024.

[7] M. López de Prado. *Advances in Financial Machine Learning*. John Wiley & Sons, 2018.

[8] Z. Zhang, S. Zohren, and S. Roberts. DeepLOB: Deep convolutional neural networks for limit order books. *IEEE Transactions on Signal Processing*, 67(11):3001–3012, 2019.